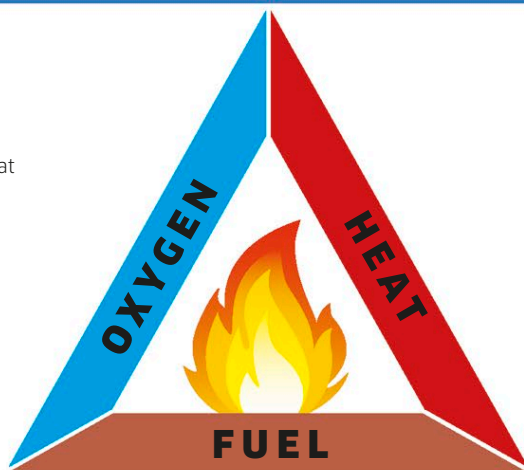


Heat and light

Combustion releases energy in the form of heat and light. Although it can feel very hot at the top of the flame, the hottest part of a flame is the blue area near its base.

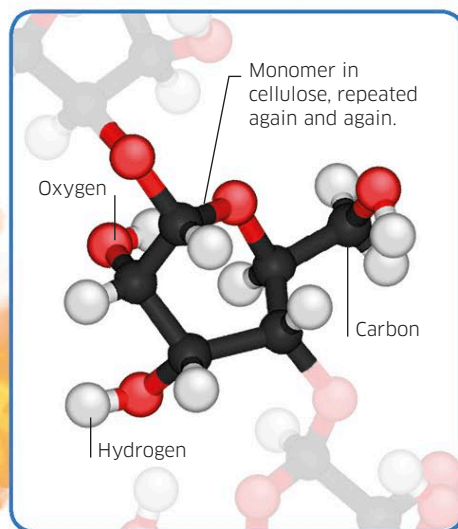


Combustion triangle

These three ingredients—fuel, oxygen, and heat—are all essential for combustion. Removing any one of them will extinguish a fire.

Fuel: firewood

Wood contains a material called cellulose. It consists of long molecules known as polymers (see p.45). Each polymer is made of a chain of smaller identical parts, called monomers. Each monomer in cellulose has six carbon atoms, ten hydrogen atoms, and five oxygen atoms, so its formula is $C_6H_{10}O_5$.

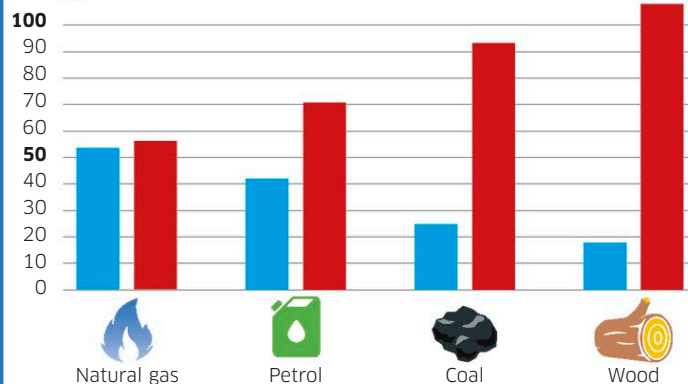


Fuel efficiency and the environment

Different fuels release different amounts of energy. They also produce different amounts of carbon dioxide when they burn. Wood is least efficient and produces the most carbon dioxide, which makes it the least environmentally friendly fuel.

Energy values of different fuels

■ Energy content (kJ per gram of fuel)
■ Quantity (mg) of carbon dioxide released per kJ of energy



Fireworks

Fireworks shoot up in the air and explode into colorful displays thanks to combustion. The fuel used is charcoal, mixed with oxidizers (compounds providing oxygen) and other agents. The colors come from different metal salts.



3 Display

Released from the head into the sky, each little "star" explodes to reveal its particular color.

2 Explosion

The next reaction happens when the fire reaches the section filled with explosives and little "stars" of metal salts.

1 Lift-off

A lit fuse reaches the lift charge and sets off the first combustion. This propels the rocket high into the sky.

